



Single nucleotide polymorphisms and penicillin non-susceptibility among invasive *Streptococcus pneumoniae* from Vietnam and India: Insights from a comparative genomics study

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ABSTRACT

Objectives: *Streptococcus pneumoniae* (*S. pneumoniae*) is a significant cause of global morbidity and mortality across all age groups. Antimicrobial resistance (AMR), particularly penicillin resistance, is a significant treatment challenge. The study investigates AMR patterns among *S. pneumoniae* isolates from Vietnam, a country with high penicillin non-susceptibility, and India which in comparison has lower penicillin non-susceptibility. The present study focuses on penicillin resistance-associated genes and single nucleotide polymorphisms (SNPs), that contribute to pathogenicity.

Methods: Invasive *S. pneumoniae* belonging to serotypes 23F ($n = 13$) and 19F ($n = 20$) genome sequences from Vietnam ($n = 13$) and India ($n = 20$) were retrieved and annotated to identify core genes. These genes were screened for antimicrobial resistance using the database. The genome sequences were mapped to a reference genome to detect the gene variants and were analysed for identifying specific mutations that contribute to pathogenicity.

Results: Annotations identified *pbp1a*, *pbp2b*, and *pbp2x* (*pbpX*) as relevant AMR genes. In Vietnam, *pbpX* was a core gene, with 41 SNPs detected, of which 7 were deleterious mutations contributing to penicillin resistance. In contrast, *pbpX* was completely absent in the Indian genomes analysed.

Conclusions: The current study highlights the genetic basis of penicillin non-susceptibility among invasive *S. pneumoniae* serotypes 23F and 19F and focuses on regional variations in resistance between India and Vietnam. The *pbpX* was identified as a core gene among penicillin-resistant pneumococci in Vietnam. In India, the absence of *pbpX* highlights genomic diversity, distinct from the uniformity observed in Vietnamese genomes. The study further predicts that deleterious SNPs are associated with pathogenicity.

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1. Introduction

Infections caused by *Streptococcus pneumoniae* (*S. pneumoniae*) pose a significant burden due to high morbidity and mortality, especially among children under 5 y of age and the elderly > 65 y [1]. Pneumococcal diseases range from noninvasive infections, like otitis media and sinusitis, to serious invasive infections, such as meningitis, bacteraemia, and pneumonia, highlighting their clinical importance. The capsule composition of *S. pneumoniae* determines

serotype classification and is associated with the invasiveness and pathogenesis of this organism [2]. Among all known pneumococcal serotypes, 23F, 19F, 6B, and 14 are commonly associated with invasive pneumococcal diseases and are major contributors to the high prevalence of β -lactam resistance. Among these serotypes, 23F and 19F have shown high resistance to penicillin antibiotics in India and Vietnam, respectively, and are important constituents in the currently licensed pneumococcal vaccines [3,4]. Although vaccination and antibiotic therapy provide some protection, the emergence of new pneumococcal serotypes and increased antibiotic resistance affect the efficacy of prevention and treatment. *S. pneumoniae* can coexist with other microorganisms in the upper respiratory tract without causing any harm. Upon transmission to other regions, this harmless colonization can trigger invasive dis-

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eases like pneumonia, meningitis, and sepsis. As it is present in the nasopharynx, it is a critical reservoir for pneumococcal infection within the community [5].

Among 5.8 million deaths annually, 318 000 are attributable to pneumococcal infections, which is nearly 10% of the deaths occurring globally. The burden of invasive pneumococcal diseases is particularly high in Southeast Asia, where in ca. 48% of all cases and 28% mortality are due to pneumococcal diseases [6]. *S. pneumoniae* has evolved to become resistant to penicillin and other antibiotics, such as macrolides and cephalosporins, which have emerged rapidly worldwide. Antibiotic resistance causes a public health burden, ca. 4.59 million deaths globally were reported for bacterial antimicrobial resistance (AMR), of which 1.27 million are directly attributed to antibiotic resistance [7]. Additionally, *S. pneumoniae* was the leading pathogen responsible for the global disability-adjusted life years associated with antibiotic resistance [8].

The antibiotic and resistance patterns in Southeast Asian countries, such as India and Vietnam, are of significant public health interest. The Survey of Antimicrobial Resistance (SOAR) study from Vietnam reported an increase *S. pneumoniae* resistance to penicillin, amoxicillin-clavulanic acid, cefaclor, and cefuroxime between 2011 and 2019, with particularly higher resistance in the 2016–2018 period [9]. In rural regions of Vietnam, a study of resistance in preschool children reported that multidrug resistance increased from 31% to 80% over a 15-y period (1999–2014) [10]. In India, resistance to *pneumococci* has increased over the years. Sing et al. observed that 7 studies in children below 5 y of age reported 10% incidence of resistance during the years 2009 to 2016 [3]. Our study focuses on penicillin non-susceptible invasive *S. pneumoniae* serotypes 23F and 19F, which are also important vaccine serotypes with the objective to identify the significant cause of penicillin non-susceptibility. The genes and their mutations, along with the significant genomic factors contributing to the resistance, were analysed for better understand the country-specific dynamics of penicillin resistance.

2. Methods and materials

2.1. Genome retrieval

The genomes of *S. pneumoniae* of Vietnamese origin were obtained from the PubMLST database, with country-specific criteria applied to identify relevant samples [11]. For India, genomic data were retrieved from the Global Pneumococcal Sequencing (GPS) project [12], as there was insufficient information on PubMLST for Indian metadata. The final dataset consisted of 20 genomes from India (18 were of serotype 19F, and 2 were of serotype 23F) and 13 from Vietnam (2 were of serotype 19F, and 11 were of serotype 23F). All sequences were filtered by isolation source and clinical condition to ensure consistency in the analysis.

2.2. Genome annotation

Genome annotation was performed using Prokka version 1.14.6 [13], a widely used automated annotation pipeline for prokaryotic genomes. Prokka utilizes multiple external tools to predict and annotate various genomic features. The input for annotation was pre-assembled genomic DNA sequences provided in FASTA format [14]. Prokka is prodigious for identifying protein coding. Moreover, it identifies other genomic features, namely repeat regions and insertion sequences. It determines the coordinates of genomic features within the contigs, providing comprehensive and reliable annotation results [15,16].

2.3. Antimicrobial resistant genes

AMR genes were identified using AMRFinderPlus [17], a highly accurate tool for detecting AMR genes, point mutations, and virulence factors. AMRFinderPlus combines protein and nucleotide sequence analysis, providing results for both formats. The tools utilize BLAST cutoffs, which are manually curated to a strict threshold of > 90%, ensuring high accuracy in prediction. The analysis used protein and nucleotide sequences obtained from Prokka annotated results. The general feature format file, which includes genomic location information, was used in conjunction with FASTA nucleic acid and FASTA amino acid file formats for AMR gene identification.

2.4. Pan-genome analysis

To perform the pan-genome analysis of *S. pneumoniae*, we used Roary version 3.13.0 [18], a tool designed for the pangenome analysis of prokaryotic genomes. It constructs the pan-genome by identifying core and accessory genes across all genomes. It uses an integrated PRANK version 170427 to construct the core genes (present in all strains) and accessory genes (present in some strains) [19,20]. For the multiple sequence alignment program, the 100% BLAST identity threshold was applied. The analysis was conducted with a clustering sensitivity threshold of 90% to identify the distribution of core, softcore, shell, and cloud genes. We used a Python script to calculate the statistical comparison and a Venn diagram to analyse the total presence and absence of the genes. It also provides an overview of the shared and unique genes present across human genomes.

2.5. Phylogenetic analysis

Randomized Axelerated Maximum Likelihood version 8 (RAXML) is a program used for the phylogenetic analysis of datasets under Maximum Likelihood (ML) methods. RAXML includes a rapid bootstrap algorithm that significantly reduces computation time. It gives a fast ML tree search algorithm with optimal likelihood scores [21]. The input for RAXML is the core gene alignment file generated by Roary. The resulting phylogenetic tree file was subsequently uploaded to the Interactive Tree of Life (iTOL) version 6 for visualization. The parameters of iTOL were customized to match the visual integrity between the genomes [22,23].

2.6. Variant calling and single nucleotide polymorphism identification

The reference genome of *S. pneumoniae* (Assembly ID: NCTC7465) was indexed using Burrows-Wheeler Aligner (BWA), which enables the efficient alignment of the sequences. The sequence was first indexed and then mapped against the genomes of Vietnam, and then aligned to the reference genome using the BWA maximal exact matches algorithm, producing a Sequence Alignment Map file containing the alignment information. The Sequence Alignment Map file was converted into a Binary Alignment Map (BAM) file using Samtools, which was then sorted to organize the alignments by their genomic coordinates. The sorted BAM file was indexed to enable rapid access to specific regions within the dataset. Variant calling was performed using BCFtools, where the first pileup of the aligned reads against the reference genome was generated. Variants were observed from the genome, then compared and indexed for efficient querying. To ensure high confidence, variant calls were applied for the filtering criteria using VCFtools, selecting variants based on quality metrics of the default minimum quality score threshold. Following variant identification, we visualized the variants using the Integrative Genomics Viewers

Table 1
Metadata and phenotypic data of Vietnamese *S. pneumoniae* genomes retrieved from the PubMLST database.

ID	Source	Year	Serotype	Clinical conditions	MIC for penicillin (mg/L)
1573	Blood	2000	23F	Bacteraemia	2
29868	Blood	1997	23F	Bacteraemia	0.75
29869	Cerebrospinal fluid	2000	23F	Meningitis	1.5
29870	Blood	1998	23F	Bacteraemia	0.125
29871	Blood	1999	23F	Meningitis	1
29872	Blood	1997	23F	Bacteraemia	1
29873	Blood	1997	23F	Bacteraemia	0.25
29874	Blood	1997	23F	Bacteraemia	1
29875	Blood	1999	23F	Bacteraemia	1.5
30623	Sputum	2001	19F	Pneumonia	0.5
30624	Blood	1999	19F	Bacteraemia	0.5
142895	Cerebrospinal fluid	1999	23F	Meningitis	1.5
29876	Blood	1997	23F	Bacteraemia	2

MIC, minimum inhibitory concentration.

(IGVs). This allows for reviewing read alignments and validating the variant calls within their genomic context, providing a comprehensive view of the identified variants and their supporting sequence data.

2.7. Variant annotations and single nucleotide polymorphism profiling

To understand the effects of variants in the genome sequence, we performed variant annotation [24]. Once the variant calling files (VCFs) were obtained from mapping, the SnpEff tool was used to annotate variants based on their genomic locations and predict their coding impacts. Annotated genomic locations include intronic, untranslated regions, upstream, downstream, splice sites, and intergenic regions. In addition, the coding effects shown are synonymous or non-synonymous amino acid replacement, start codon gains or losses, stop codon gains or losses, or frameshift mutation are predicted. Missense mutations involve amino acid changes critical for understanding the functional impacts of the genome hence, we proceeded with missense mutations [25].

2.8. Prediction of pathogenicity of single nucleotide polymorphisms

Single nucleotide polymorphisms (SNPs) were analysed to predict the pathogenicity using different bioinformatics tools. Sorting intolerant from tolerant (SIFT), which uses sequence homology for interpreting the effects of amino acid substitution to identify the tolerant or deleterious SNPs, was used with a prediction probability of $p > 0.05$ [26]. Protein analysis through the evolutionary relationship, the (PANTHER) program, was used to classify proteins based on evolutionary relationships, molecular function, and interactions with other proteins. The SNPs with a prediction probability of $p > 0.05$ are considered deleterious [27]. Polymorphism phenotyping version 2 (PolyPhen-2) predicts the functional effect of amino acid substitutions on proteins [28]. The protein variation effect analyser (PROVEAN) webserver provides a prediction based on the functional effect of single or multiple amino acid substitutions using support vector machines. It uses a cutoff scores that was set to -2.5 for highly balanced accuracy [29]. The synonymous non-synonymous analysis program (SNAP) is a method that helps in predicting the functional effects of amino acid substitutions with a probability of >0.5 was considered [30]. Predict SNP is the consensus SNP classifier that has six prediction programs (MAPPP, PhD-SNP, PholyPhen-1, PholyPhen-2, SIFT, and SNAP) to predict the deleterious mutations [31]. Meta SNP was used to predict the single point mutation variations that can be classified based on the disease-associated or polymorphism (neutral). It is a consensus of four prediction programs (PANTHER, PhD-SNP, SIFT, and SNAP) [32,33]. These were used to ensure the accuracy of the results, and SNPs that are vital for causing resistance and expected

to be deleterious across most of the nine programs have been considered [34,35].

3. Results

3.1. Retrieval of genomes

The genomes were filtered based on sources (blood and cerebrospinal fluid), serotypes (23F and 19F), and clinical conditions (pneumonia, bacteraemia and meningitis). The 13 genomic sequences from Vietnam obtained from the PubMLST had penicillin minimum inhibitory concentration (MIC) values in the range of 0.125 to 2 µg/mL. These isolates and their corresponding MIC values and other relevant information are summarised in Table 1. The initial dataset of Indian genomes from the GPS project comprised 682 samples, which were filtered based on the source, serotype, and clinical conditions. This screening process resulted in a refined dataset of 20 samples, providing a focused representation for further analysis. The penicillin MIC values of these samples ranged from 0.125 to >8 µg/mL, one isolate having a penicillin susceptible MIC value of 0.06 mg/L was included for comparative purposes. Details of the selected samples are presented in Table 2.

3.2. Genome annotation and AMR genes

Using Prokka, all genome samples were annotated, and the result predicted the number of contigs, CDS, rRNAs, and tRNAs across the datasets. In the Vietnam samples, the number of CDS ranged from 1495 to 1974, rRNAs between 3 and 6, and tRNAs between 16 and 35. For samples belonging to India, the CDS ranged between 2014 and 2123, rRNAs between 5 and 6, and tRNAs from 44 to 74 (Table 3). The AMR genes were identified using AMRFinderPlus. In India and Vietnam, there were a total of 113 and 64 AMR genes showing resistance to antibiotic classes, including β -lactams, tetracyclines, aminoglycosides, and macrolides, respectively. Our primary focus was on β -lactams, particularly emphasizing *pbps*. Among the 20 sequences of isolates from India showing resistance to β -lactams, *pbp2x* was detected in 11 sequences, *pbp2b* in 19 sequences, and *pbp1a* in 4 sequences. Similarly, among the 13 sequences from Vietnam showing resistance to β -lactams, *pbp1a* was detected in 11 sequences, *pbp2b* in 13 sequences, and *pbp2x* in 10 sequences (Table 4).

3.3. Pan-genome analysis

A pan-genome profiling was performed using the annotated genes from the genomes of Vietnam and India for *S. pneumoniae*. For the Indian genomes, a total of 3015 gene clusters were found, of which 1545 (99%) were identified as core genes (genes shared

Table 2Metadata and phenotypic data of Indian genomes *S. pneumoniae* retrieved from the Global Pneumococcal Sequencing project.

ID	Source	Year	Serotype	Clinical conditions	MIC for penicillin (mg/L)
GPS_IN_CMC_28	Blood	2017	23F	Pneumonia	0.125
GPS_IN_CMC_58	Blood	2017	19F	Pneumonia	1
GPS_IN_CMC_59	Blood	2017	19F	Pneumonia	1
GPS_IN_CS139	Cerebrospinal fluid	2016	19F	Meningitis	1
GPS_IN_G08	Cerebrospinal fluid	2014	19F	Meningitis	0.5
GPS_IN_G126	Blood	2018	23F	Bacteraemia	2
GPS_IN_G25	Blood	2014	19F	Bacteraemia	4
GPS_IN_G50	Blood	2015	19F	Bacteraemia	4
GPS_IN_G65	Blood	2015	19F	Bacteraemia	4
GPS_IN_G81	Blood	2016	19F	Bacteraemia	4
GPS_IN_G82	Blood	2016	19F	Bacteraemia	4
GPS_IN_P100	Cerebrospinal fluid	2016	19F	Meningitis	> 8
GPS_IN_P105	Blood	2016	19F	Bacteraemia	2
GPS_IN_P8	Blood	2013	19F	Bacteraemia	≤ 0.06
GPS_IN_P9	Blood	2013	19F	Bacteraemia	4
GPS_IN_P94	Blood	2016	19F	Bacteraemia	2
GPS_IN_P96	Blood	2016	19F	Bacteraemia	4
GPS_IN_P97	Blood	2016	19F	Bacteraemia	4
GPS_IN_P98	Blood	2016	19F	Bacteraemia	> 8
GPS_IN_TVM_3	Cerebrospinal fluid	2012	19F	Meningitis	2

Table 3

Gene prediction statistics for genomes.

ID	Contigs	Bases	CDS	rRNA	tRNA
Vietnam					
142895	1611	1,692,894	1495	5	16
1573	534	2,028,233	1899	3	38
29868	772	2,011,988	1871	4	35
29869	1158	2,004,759	1793	3	30
29870	1319	1,973,122	1671	4	29
29871	467	2,069,131	1911	6	33
29872	358	2,054,555	1920	3	32
29873	809	1,911,265	1757	3	32
29874	282	2,032,389	1927	3	32
29875	474	2,006,273	1880	3	32
29876	568	2,006,824	1877	3	32
30623	107	2,036,625	1974	3	33
30624	131	2,038,530	1972	3	32
India					
GPS_IN_CMC_28	19	2,089,398	2057	5	46
GPS_IN_CMC_58	20	2,087,912	2069	6	47
GPS_IN_CMC_59	19	2,088,527	2068	5	48
GPS_IN_CS139	35	2,102,655	2077	6	47
GPS_IN_G08	25	2,121,004	2103	6	46
GPS_IN_G126	32	2,155,399	2120	5	47
GPS_IN_G25	31	2,076,277	2061	6	46
GPS_IN_G50	25	2,072,945	2063	5	46
GPS_IN_G65	36	2,035,726	2014	5	47
GPS_IN_G81	27	2,074,816	2064	5	45
GPS_IN_G82	27	2,073,884	2062	6	46
GPS_IN_P100	32	2,069,232	2048	5	45
GPS_IN_P105	34	2,072,262	2055	6	46
GPS_IN_P8	23	2,084,228	2070	5	46
GPS_IN_P94	35	2,072,031	2055	6	44
GPS_IN_P96	30	2,058,806	2036	5	45
GPS_IN_P97	27	2,111,008	2123	5	48
GPS_IN_P98	30	2,071,599	2060	5	46
GPS_IN_P9	30	2,075,273	2059	5	55
GPS_IN_TVM_3	26	2,074,665	2063	5	74

GPS, Global Pneumococcal Sequencing.

across all genomes). Additionally, 116 softcore genes (95%) were identified in 19 strains, 781 shell genes (15%) in 3 strains, and 603 cloud genes (not more than 3 strains) were identified (Figs. 1a–f). In the Vietnamese genomes, there were a total of 2763 gene clusters, with 975 (99%) core genes, and no softcore genes were identified, whereas 1413 shell genes were observed in 1 strain, and 381 cloud genes in less than 1 strain were observed (Figs. 2a–f). In terms of *pbp* genes, the *pbpX* gene was consistently found

within the gene clusters. However, the *pbpX* gene was not considered a core gene in Indian genomes, as it was observed only in 17 out of 20 genomes. Among the three genomes lacking *pbp* genes, one revealed penicillin susceptibility (ID: GPS_IN_P8), whereas the other two exhibited non-susceptibility (IDs: GPS_IN_CMC_28 and GPS_IN_G08). In Vietnam, it was observed that all 13 genomes had *pbpX* as the core gene, further focusing on the Vietnamese dataset. The comparison of the shared and unique genes and the *pbpX* pres-

Table 4
Occurrence of *pbp* genes from AMR gene profiling.

Element name	Element symbol	Frequency (n = 20)	Antibiotic resistance ontology (ARO) term	Detection criteria	Drug CLASS	% Identity of matching region	% Length of reference sequence
Vietnam							
Streptococcus pneumoniae beta-lactam resistant PBP1A	pbp1a	11	Penicillin-binding protein PBP1A1	AMR	BETA-LACTAM	92.9318	100
Streptococcus pneumoniae beta-lactam resistant PBP2B	pbp2b	13	Penicillin-binding protein PBP2B2	AMR	BETA-LACTAM	97.0054	100
Streptococcus pneumoniae beta-lactam resistant PBP2X	pbp2x	10	Penicillin-binding protein PBP2×23	AMR	BETA-LACTAM	90.65	100
India							
Streptococcus pneumoniae beta-lactam resistant PBP2X	pbp2x	11	Penicillin-binding protein PBP2×2	AMR	BETA-LACTAM	92.0509	100
Streptococcus pneumoniae beta-lactam resistant PBP2B	pbp2b	19	Penicillin-binding protein PBP2B2	AMR	BETA-LACTAM	95.2963	100
Streptococcus pneumoniae beta-lactam resistant PBP1A	pbp1a	4	Penicillin-binding protein PBP1A1	AMR	BETA-LACTAM	91.17	100

AMR, antimicrobial resistance.

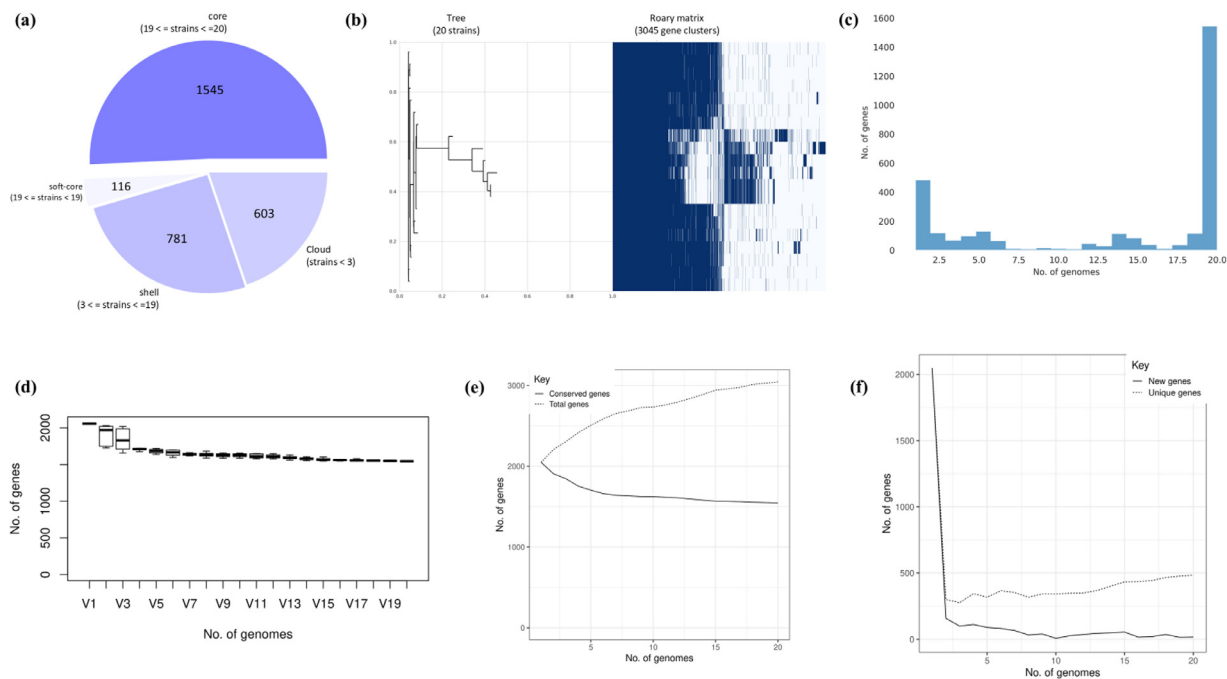


Fig. 1. Pan-genome profile of the Indian genomes figure (a) pie chart representing gene clusters of the pan genomes, (b) present/absent matrix with phylogenetic clustering, (c) gene frequency plot, (d) Box plot showing the number of conserved genomes across the pan-genome, and (e) graph showing comparison of conserved genes to total genes, (f) graph showing comparisons of new genes and unique genes.

ence is shown in Figs. 3a and b. All gene details are given in Supplementary Table S1.

3.4. Phylogenetic analysis

Phylogenetic relationships among the genomes were performed using the ML method. A starting tree that included all the core gene alignment sequences was utilized for the analysis, while the substitution model was set to general time reversible and a gamma distribution was assigned to estimate the distribution of the rates heterogeneity. The core genes in the Vietnamese genomes showed a high level of heterogeneity resulting in 772 different alignment patterns. Similarly, the Indian genomes showed 71 unique alignment patterns in the gamma model. The trees were visualised with iTOL, which represented the evolutionary relationships. The tree structure reveals the similarities of the core genes across the Vietnamese and Indian genomes, as represented in Figs. 4a and b.

3.5. Determination of variants

The high-quality trimmed reads were aligned to the reference genome. The total average number of reads across all samples was 741.69, with an average of 644.69 (86%) reads mapping to the reference genome. The resulting BAM files were sorted and indexed, ensuring efficient access for variant calling using IGV visualisation. The IGV visualisation of the filtered VCF files revealed 20 254 variants across the genome.

3.6. Variant annotation

The variant statistics for each sample are summarized in Table 5. Among deletions (DELs), insertions (INS), and SNPs identified from the genomes, more than 99% were dominated by SNPs across all the samples. Missense mutations contribute to an aver-

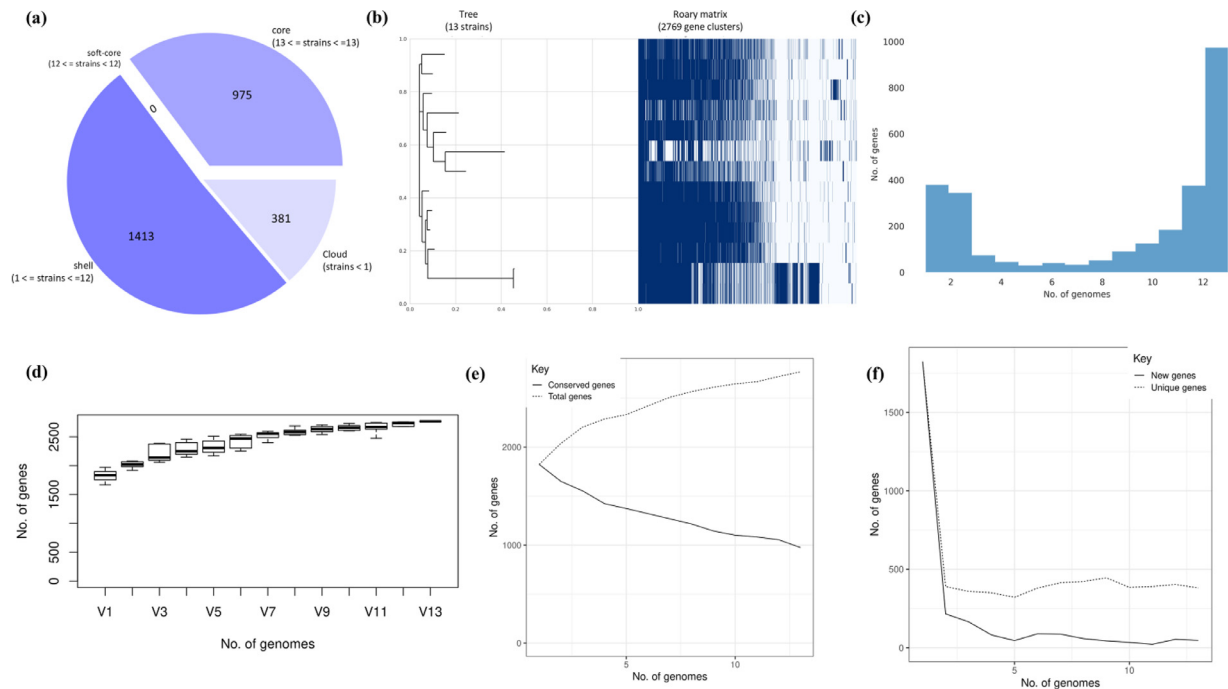


Fig. 2. Pan-genome profile of the Vietnam genomes figure (a) pie chart representing gene clusters of the pan genomes, (b) present/absent matrix with phylogenetic clustering, (c) gene frequency plot, (d) Box plot showing the number of conserved genomes across the pan-genome, and (e) graph showing comparison of conserved genes to total genes, (f) graph showing comparisons of new genes and unique genes.

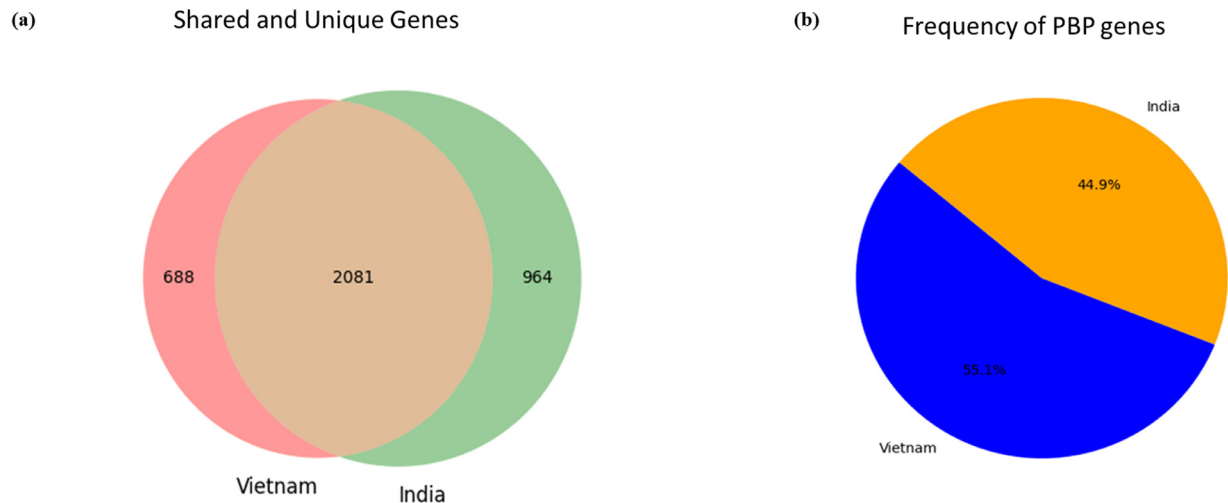


Fig. 3. (a) Venn diagram depicting shared and unique genes between Vietnam and India (b) Pie chart depicting the frequency of *pbp* genes in Vietnamese and Indian genomes.

Table 5
Variant annotation statistics of identified SNPs from the Vietnam genomes.

ID	DEL	INS	SNP	HIGH	Low	Moderate	Modifier	Missense	Nonsense	Silent	Downstream	Exon	Intron	Upstream
1573	0.35%	0.51 %	99.14 %	0.08 %	5.07 %	2.37 %	92.48 %	31.86 %	0.43 %	67.71 %	41.43 %	7.51 %	8.47 %	42.58 %
29868	0.40 %	0.46 %	99.13 %	0.08 %	5.10 %	2.34 %	92.49 %	31.44 %	0.38 %	68.18 %	41.19 %	7.50 %	8.48 %	42.82 %
29869	0.31 %	0.42 %	99.27 %	0.08 %	5.14 %	2.33 %	92.46 %	31.25 %	0.38 %	68.36 %	41.14 %	7.54 %	8.47 %	42.84 %
29870	0.35 %	0.39 %	99.25 %	0.08 %	5.20 %	2.34 %	92.38 %	31.04 %	0.42 %	68.53 %	41.13 %	7.61 %	8.46 %	42.79 %
29871	0.41 %	0.46 %	99.13 %	0.09 %	4.99 %	2.35 %	92.58 %	32.02 %	0.48 %	67.50 %	41.25 %	7.42 %	8.46 %	42.86 %
29872	0.35 %	0.41 %	99.24 %	0.08 %	5.02 %	2.34 %	92.56 %	31.84 %	0.39 %	67.76 %	41.32 %	7.43 %	8.47 %	42.78 %
29873	0.33 %	0.39 %	99.28 %	0.08 %	5.19 %	2.36 %	92.38 %	31.25 %	0.39 %	68.36 %	41.35 %	7.62 %	8.48 %	42.55 %
29874	0.38 %	0.49 %	99.13 %	0.09 %	4.99 %	2.35 %	92.57 %	32.02 %	0.49 %	67.50 %	41.38 %	7.42 %	8.46 %	42.73 %
29875	0.35 %	0.43 %	99.22 %	0.08 %	5.03 %	2.35 %	92.54 %	31.92 %	0.44 %	67.63 %	41.56 %	7.46 %	8.47 %	42.50 %
30623	0.42 %	0.54 %	99.04 %	0.09 %	4.90 %	2.37 %	92.65 %	32.53 %	0.51 %	66.96 %	41.77 %	7.35 %	8.44 %	42.43 %
30624	0.44 %	0.55 %	99.01 %	0.09 %	4.88 %	2.39 %	92.64 %	32.83 %	0.51 %	66.66 %	41.81 %	7.35 %	8.44 %	42.40 %
142895	0.31 %	0.31 %	99.37 %	0.07 %	5.41 %	2.39 %	92.14 %	30.68 %	0.28 %	69.04 %	40.87 %	7.86 %	8.44 %	42.83 %
29876	0.33 %	0.41 %	99.26 %	0.08 %	5.07 %	2.35 %	92.51 %	31.67 %	0.42 %	67.90 %	41.41 %	7.49 %	8.47 %	42.62 %

DEL, deletion; INS, insertion; SNP, single nucleotide polymorphism.

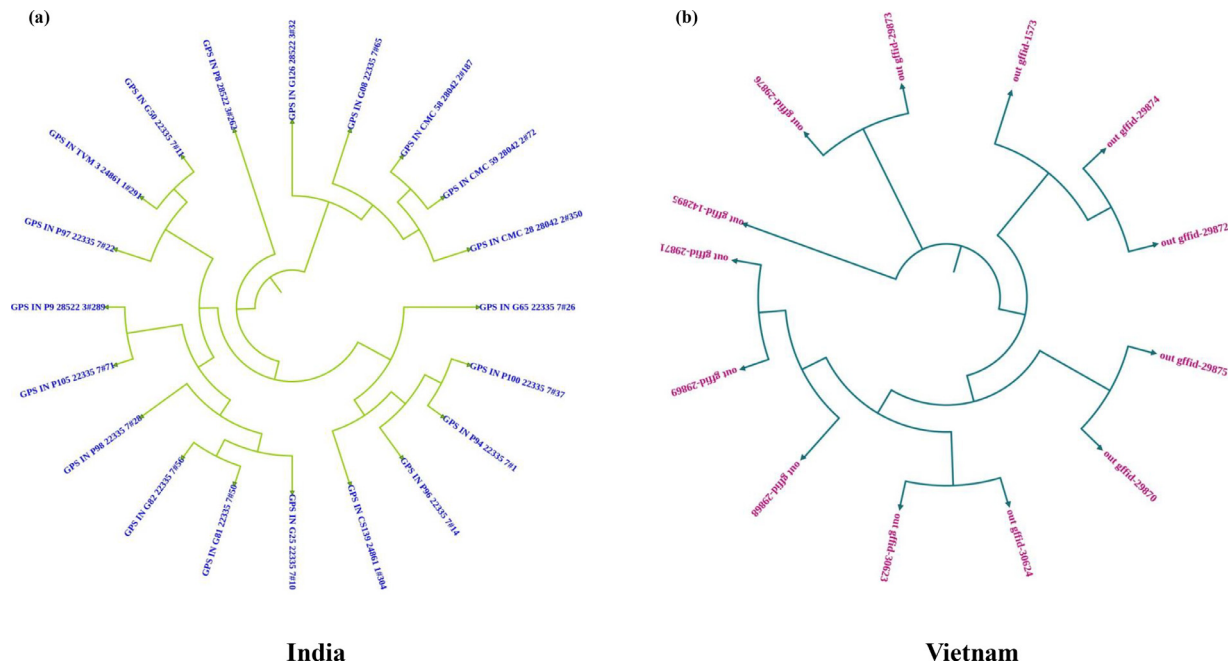


Fig. 4. Phylogenetic trees based on the core gene alignment (a) India and (b) Vietnam.

Table 6
Variant annotation statistics of SNPs in the *pbp2x* gene from Vietnam genomes.

ID	VI Low	VI Moderate	VI Modifier	VE downstream gene variant	VE missense variant	VE synonymous variant	VE upstream gene variant
1573	205	85	125	105	85	205	20
29868	197	85	105	85	85	197	20
29869	205	85	104	84	85	205	20
29870	205	85	105	85	85	205	20
29871	205	85	110	90	85	205	20
29872	205	85	127	107	85	205	20
29873	205	85	110	90	85	205	20
29874	205	85	126	106	85	205	20
29875	205	85	127	107	85	205	20
30623	237	103	326	117	103	237	209
30624	237	103	327	118	103	237	209
142895	205	87	71	54	87	205	17
29876	205	85	124	104	85	205	20

SNP, single nucleotide polymorphism.

age of 30% of the variants, whereas synonymous (silent) mutations account for around 67%. Other variant effects, such as those categorized as upstream, downstream, exon, and intron, also displayed consistent trends across the samples. The total number of variants observed across all samples ranged from 16 000 to 19 500. Focusing on the *pbpX* (*pbp2x*) gene, we found an average of 205 low-impact, 85 moderate-impact, and 125 modifier variants per sample. In the *pbpX* gene from all samples, 85 were missense mutations, 20 were upstream variants, and 205 were synonymous mutations, indicating a highly conserved region across the samples. To refine the analysis, the missense mutations were further filtered to remove redundancies, resulting in 41 mutations across all the samples. These unique mutations were used for subsequent SNP analysis. These findings are depicted in Table 6. The tables summarizing the overall variant profile are provided in Supplementary Table S2.

3.7. Pathogenicity of the SNPs

The pathogenicity of the identified variants was estimated using multiple computational tools to understand the potential impact of mutations in *pbpX*. The MAPP software predicted 11 deleterious and 24 neutral effects; SNAP prediction showed 10 deleterious and 31 neutral; SIFT prediction had 13 deleterious and 28 neutral;

PolyPhen prediction found 5 deleterious and 36 neutral; Phd-SNP prediction showed 19 deleterious and 22 neutral; PANTHER had 11 deleterious and 27 neutral; Predict-SNP had shown 5 deleterious and 36 neutrals; Meta-SNP showed 8 deleterious and 33 neutral, and for PROVEAN, 5 deleterious and 36 neutral effects were observed. Among the 41 unique SNPs analysed, 7 mutations (T338A, G355R, V358F, I371T, G382S, L510P, and S531N) were consistently predicted to have deleterious effects across most of the software, suggesting their potential role in pathogenicity. The software prediction of the SNP pathogenicity is summarized in Table 7.

4. Discussion

The prevalence rate of resistance *S. pneumoniae* toward β -lactams has been increasing and is among the most important AMR trends tracked globally [36,37]. Previous studies have demonstrated increased association of penicillin resistance with serotypes 19F and 23F, necessitating investigation of its underlying genetic determinants [38]. In 2018, Kulkarni and her team reported that 14% of penicillin resistance was observed in India, which later increased to 19% in 2019 [39]. According to the report provided by Quang et al. during the years 2020 to 2021, penicillin has shown an increased resistance of 98.99% in Vietnam [40]. Thus, it is essen-

Table 7
Pathogenicity predictions of the *pbp2x* mutations.

Mutations	MAPP prediction	PhD-SNP prediction	PolyPhen prediction	SIFT prediction	SNAP prediction	PANTHER prediction	PredictSNP prediction	Meta-SNP	PROVEAN
D311N	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
T338A*	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Deleterious	Neutral	Deleterious	Deleterious
M343T	Neutral	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
A346S	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
A347S	Deleterious	Deleterious	Neutral	Neutral	Neutral	Deleterious	Neutral	Neutral	Neutral
G355R*	Deleterious	Deleterious	Deleterious	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Neutral
V358D	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Neutral	Neutral	Deleterious	Neutral
V358F*	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Deleterious	Deleterious	Neutral	Neutral
L364F	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
I371T*	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Deleterious	Neutral	Deleterious	Deleterious
E378D	Unknown	Deleterious	Neutral	Neutral	Neutral	Unknown	Neutral	Neutral	Neutral
G382A	Unknown	Deleterious	Neutral	Deleterious	Deleterious	Neutral	Neutral	Deleterious	Neutral
G382S*	Unknown	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Deleterious	Deleterious	Neutral
R384G	Neutral	Neutral	Neutral	Neutral	Neutral	Deleterious	Neutral	Neutral	Neutral
S389F	Neutral	Deleterious	Deleterious	Neutral	Neutral	Deleterious	Neutral	Neutral	Deleterious
T401S	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
L413F	Neutral	Neutral	Neutral	Deleterious	Neutral	Deleterious	Neutral	Neutral	Neutral
N417K	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
N444S	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
I462L	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
P486T	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Deleterious
D488N	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
T490S	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
A491V	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
D506E	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
L510I	Neutral	Neutral	Neutral	Deleterious	Neutral	Deleterious	Neutral	Neutral	Neutral
L510P*	Deleterious	Deleterious	Neutral	Deleterious	Deleterious	Deleterious	Deleterious	Deleterious	Neutral
T513N	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
N514H	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
V516I	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
V523L	Unknown	Neutral	Neutral	Neutral	Neutral	Unknown	Neutral	Neutral	Neutral
S531N*	Unknown	Deleterious	Deleterious	Deleterious	Deleterious	Deleterious	Deleterious	Deleterious	Neutral
S531C	Unknown	Neutral	Neutral	Deleterious	Deleterious	Neutral	Neutral	Neutral	Neutral
T536I	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
V537I	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
L546V	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
D567N	Neutral	Neutral	Neutral	Neutral	Neutral	Unknown	Neutral	Neutral	Neutral
A572V	Deleterious	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
S574T	Deleterious	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
S576N	Deleterious	Deleterious	Neutral	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral
N605T	Deleterious	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Deleterious

* Represents the frequently occurring SNPs, and Bold represents the Deleterious nature of mutations. PROVEAN, protein variation effect analyser; SIFT, sorting intolerant from tolerant; SNAP, synonymous non-synonymous analysis program; SNP, single nucleotide polymorphism.

tial to delve deep into understanding the resistance mechanisms developed against β -lactams notably penicillin in these two countries. In this study, we focused on understanding the resistance dynamics in both countries, unveiling the existing and novel mutations leading to penicillin resistance. Additionally, the analysis in the current work extended toward identifying significant mutations and evaluating their pathogenicity, advancing comparative genomic strategies through variant annotation and SNP profiling. The resistance patterns and the underlying genetic determinants playing key roles in the resistance were specifically reported in the Vietnamese genomes.

We have included the phenotypic and genotypic information of *S. pneumoniae* from India and Vietnam, focusing on AMR mechanisms and their genomic validation. Sequence data from the PubMLST and the GPS project were utilized, comprising 13 sequences from Vietnam and 20 from India, selected based on various parameters. MIC values collected from various regions provide critical information on penicillin resistance. The genome annotation identifies multiple AMR genes, with significant priority on *pbps*, particularly *pbp2x*, *pbp2b*, and *pbp1a*. These *pbps* are well-reported contributors to β -lactam resistance, specifically to penicillin and related antibiotics [41,42]. Pan-genome analysis revealed that among the *pbp* genes, *pbpX* (*pbp2x*) was consistently present in the core genome of Vietnamese samples but absent in the core genome samples of India. It is observed that the ab-

sence of the *pbpX* genes in the three Indian genomes (GPS_IN_P8, GPD_IN_CMC_28, and GPS_IN_G08) correlates with the decrease in the MIC range. The presence of *pbpX* exclusively was found in the Vietnam core genomes. This suggests that *pbpX* has a critical role in the resistance mechanism, with its known contribution to β -lactam resistance [43,44]. Furthermore, in Vietnam, the serotypes 19F and 23F have shown a high level of penicillin resistance (>90%) correlated with 99.6% *pbp2x* presence [45]. To understand the resistance mechanisms, the Vietnamese sequences were aligned, and a reference genome was retrieved from the NCBI database, identifying 205 variants specific to *pbpX*. Notably, samples ID- 30623 and ID- 30624 Vietnamese samples exhibited a higher number of missense mutations, indicating a potential hotspot for variation in *pbpX*. The annotation and filtering showed 41 unique mutations, including T338A, N514H, and I371T, which were previously reported mutations associated with resistance. T338A alters the active site of *pbp2x*, reducing β -lactam binding efficacy, whereas I371T induces structural changes that further contribute to resistance [46–49].

To assess the pathogenic potential of these mutations, the SNP pathogenicity prediction identified seven mutations (T338A, G355R, V358F, I371T, G382S, L510P, and S531N) as deleterious. Among these, T338A and I371T have been reported in prior studies [46,48–50], whereas G355R, V358F, G382S, L510P, and S531N are novel mutations that have not yet been reported. These novel

mutations, predicted as deleterious by most of the tools, could represent new mechanisms contributing to β -lactam resistance. These findings highlight the need for further functional studies to elucidate the role of these novel mutations in resistance pathways. The absence of *pbpX* in the Indian core genome samples aligns with previously reported variations in resistance profiles between India and Vietnam [51]. This difference shows how regional and evolutionary pressures shape AMR gene distribution. Additionally, the identification of novel deleterious mutations suggests unique evolutionary adaptations in Vietnam, potentially driven by selective pressures from local antibiotic usage patterns and healthcare practices [52].

Understanding the resistance mechanisms associated with *pbpX* mutations requires further exploration. Functional validation through experimental studies is crucial to determine their specific contributions to the resistance. For instance, structural studies of mutated *pbpX* proteins can reveal changes in drug binding affinity. Investigating these mutations in clinical isolates will also help to clarify their prevalence and impact in diverse populations. This study highlights the efficacy of combining phenotypic data with genomic insights to uncover the molecular mechanisms underlying AMR. The identification of known and novel *pbpX* mutations not only expands our understanding of β -lactam resistance but also signifies the importance of molecular surveillance in tracking and mitigating resistant pneumococcal infections. This finding specifies the necessity of tailored region-specific AMR prevention strategies, backed by local genomic and epidemiological data.

5. Limitations

The genome sequence from Vietnam was retrieved from the older datasets due to the unavailability of recent genomic data, although higher penicillin resistance has been reported in recent years.

6. Conclusions

The present study focused on AMR mechanisms in Vietnam and India by analysing the genetic profiles of samples from both countries. Our findings reveal a significant difference in resistance patterns between the two countries. SNP analysis in the Vietnam samples identified 41 SNPs, with 7 having deleterious effects linked to bacterial pathogenicity. The findings have indicated that high penicillin resistance in Vietnam is likely due to the presence of *pbpX* as a core gene that plays a significant role in driving resistance. This highlights the genetic basis for varying resistance patterns and emphasizes the importance of geographical region-specific strategies to combat AMR in *S. pneumoniae*.

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Data availability: The genomes (datasets) analysed during the current study are available in the PubMLST (<https://pubmlst.org/>) for Vietnam, the Global Pneumococcal Sequencing project (<https://www.pneumogen.net/gps/>) for India, and the National Centre for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/>) for the reference genome.

All data supporting the findings of this study are available in the article, as well as supplementary information.

Author contributions: Literature search, data curation, and writing the original draft were performed by Avani Panickar. Study conceptualization, supervision, reviewing, and editing were conducted by Anand Manoharan. Study conceptualization and funding acquisition were done by Sudha Ramaiah.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2025.04.016.

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